

TASK 01

```
10
22
0
98
-12
54
32
17
67
36
82
22 0 98 -12 54 32 17 67 36 82
Total swaps made: 16
Total Comparisons made: 35 (Actual comparisons for worst case: 45)
Number of passes saved: 4
-12 0 17 22 32 36 54 67 82 98
```

TASK 02

```
7
5
8
2
1
9
0
3
Before Sorting:
5 8 2 1 9 0 3
Iteration 1:
key 8 is in correct position. No shift required
Iteration 2:
Key 2 required 2 number of shifts.
2 5 8 1 9 0 3
Iteration 3:
Key 1 required 3 number of shifts.
1 2 5 8 9 0 3
Iteration 4:
key 9 is in correct position. No shift required
Iteration 5:
Key 0 required 5 number of shifts.
0 1 2 5 8 9 3
Iteration 6:
Key 3 required 3 number of shifts.
0 1 2 3 5 8 9

Total shifts: 13
After Sorting:
0 1 2 3 5 8 9
```

TASK 03

```
10
5
7
2
1
0
4
8
3
9
6
Before Sorting:
5 7 2 1 0 4 8 3 9 6
Total swaps: 6
Skipped swaps: 3
Swap to Comparison Ratio: 13.3333%
Total Comparisons: 45
After Sorting:
0 1 2 3 4 5 6 7 8 9
```

TASK 04

```
9
23
29
15
19
31
7
9
5
2
Before Sorting:
23 29 15 19 31 7 9 5 2
Gap: 4
Gap Percentage: 44.4444%

Number of comparisons in gap = 4 : 6
Number of swaps in gap = 4 : 5
Gap: 2
Gap Percentage: 22.2222%

Number of comparisons in gap = 2 : 9
Number of swaps in gap = 2 : 3
Gap: 1
Gap Percentage: 11.1111%

Number of comparisons in gap = 1 : 9
Number of swaps in gap = 1 : 1
After Sorting:
2 5 7 9 15 19 23 29 31
```

TASK 05

```
9
1
2
3
4
5
6
7
8
9
Target ID? 8
LOW | MID | HIGH
  0 |  4 |  8
Remaining Search Percentage:44.4444%
Steps taken so far | theoretical max steps: 1 | 4

LOW | MID | HIGH
  5 |  6 |  8
Remaining Search Percentage:22.2222%
Steps taken so far | theoretical max steps: 2 | 4

LOW | MID | HIGH
  7 |  7 |  8
Target Index: 7
```